



Handbook of Flood Risk Management in Developing Countries

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Impacts of flooding on agriculture and food security in developing countries

Evidence from Southeastern Nigeria

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Introduction

Flooding occurs when a normally dry area is completely or partially submerged by runoff. Flooding is an intractable environmental problem that has negative impacts on agriculture and food security in developing countries. Developing countries have been noted to be predominantly affected by floods because of lack of resources, infrastructure and effective disaster management programs (D'Odorico & Rulli 2013). The international development community has recognized agriculture as being central to economic growth, food security and poverty alleviation, especially in countries where it is the main source of livelihood for the poor (World Bank 2007; FAO2011).

Agriculture enhances food security in two ways: food production and provision of livelihood sources. In Africa, about 70% of the population depends on agrarian sources of livelihood, 40% of export revenue is agriculture dependent and agriculture generates about one-third of their national income (Yaro 2004; McCusker & Carr 2006). The poorest countries in Africa have been observed to be those that depend on rain-fed subsistence agriculture for food, income and survival, and these countries are invariably the most vulnerable to any change in climate (Yaro 2004).

Climate change has been noted to represent abnormal climatic situations that influence temperature and water availability, thereby affecting agricultural production (Syaukat 2011), and would intensify the current food insecurity and hunger problems in developing countries which already contend with chronic food problems, since this has a multiplier effect on the risk of crop failure and livestock loss. Many studies indicate that the high frequency and intensity of some disasters such as floods affect agricultural production as well as food (FAO 2007; Ngho et al. 2011; Syaukat 2011).

Despite increasing global agricultural production (Ibok et al. 2014), large numbers of people remain malnourished and hungry (Ambali et al. 2015). In 2010, 925 million people were estimated to be undernourished, and about 900 million of these people reside in developing countries (FAO, 2010). According to Bashir et al. (2012), more than 70% of undernourished

people depend on agriculture directly or indirectly and live in rural areas. Similarly, IFAD (2009) estimated that about 1.2 billion people are unable to meet the most basic needs for sufficient food daily, and the largest segment of these people are the 800 million poor children, women and men who belong to indigenous populations living in rural areas, especially in developing countries, trying to make a living from subsistence agriculture. These indigenous populations rely heavily on rain-fed agriculture, which makes them vulnerable and adversely affected in times of floods.

In 2019, about 750 million globally were exposed to severe levels of food insecurity. The world has been noted not to be on track to achieving zero hunger by 2030, and over 840 million people have been projected to be affected by hunger by 2030 if this trend continues (FAO et al. 2020). It is expected that flooding will further expose these already vulnerable people to higher levels of food insecurity by reducing their nutritional status due to its negative impacts on food security and agriculture.

Food security, according to FAO (1996, 2008a), exists when there is economic and physical access to safe, sufficient and nutritious food by people at all times, thereby meeting their food preferences and dietary needs for an active and healthy life. Food security in this context consists of four pillars (food affordability, availability, utilization and stability), and these pillars must be fulfilled to realize food security objectives. Attaining a state of food security means sufficient food is available, everyone has access (economic and physical) to the food they need, access and availability are maintained over time (stability) and the food is effectively utilized. This implies that an adequate level of food consumption and good nutrition must be maintained at low risk over time (FAO 2008a); that is, it must be sustainable to achieve a state of food security. However, it has been noted that not all healthy diets are sustainable and not all sustainable diets are always healthy (FAO et al. 2020), and this is usually determined by food affordability.

Floods, like other disasters, have been noted to be linked to all four pillars of food security (Israel & Briones 2012; Ramakrishna et al. 2014; Ajaero 2017; Pacetti et al. 2017; Akukwe et al. 2018; Oskorouchi & Sousa-Poza 2021) in developing countries, as floods disrupt and damage agriculture, which is people's major source of food. According to Pacetti et al. (2017, p. 1), the inner correlations between floods and food security in developing countries are extremely relevant because they depend primarily on agriculture, which can be highly affected by extreme events, damaging their primary access to food. Oladokun and Proverbs (2016) have said flooding is one of the major culprits responsible for the high poverty level of African populations.

Flooding has been noted to negatively impact agriculture and food security by causing loss of livestock, destruction of crops, reduction in quality and quantity of food eaten, increase in price of food, destruction of roads, destruction of farm storage facilities and stream pollution (Israel & Briones 2012; Pacetti et al. 2017; Akukwe et al. 2018, 2020; Week & Wizer 2020). These negative impacts are felt mostly by households in rural areas with a paucity of social amenities such as good roads and pipe-borne water and whose primary source of livelihood is agriculture, as revealed by available literature on impacts of flooding on agriculture and food security in developing countries. This is an indication that rural dwellers residing in the hinterlands experience these impacts first-hand more than urban dwellers, whose primary occupation is not usually farming. Thus, this chapter focuses on bringing to light the areas in which floods have affected agriculture and food security and the relationship between flooding and food security with insights from southeastern Nigeria and on suggesting policy measures to minimize the impacts of flooding on agriculture and food security in developing countries.

Literature review and conceptual framework

Literature review

Over the years, studies on the impacts of natural disasters and climate have been attracting increasing attention among researchers, developers, climatologists, environmentalists and policy makers. This has, however, generated an agreement that there is indeed a relationship between floods, food security and agriculture. A number of studies have shown the many adverse events of floods resulting from climate change in addition to addressing flooding's potential effects on agriculture and food security (Parry et al. 2004; Fischer et al. 2005; Parry et al. 2005; Akukwe et al. 2020). A relationship has been observed between temperature and changing rainfall patterns that influences flooding. Several studies have indicated that raised temperature will not only change rainfall patterns but will increase the severity and frequency of harsh weather (Haines & Patz 2004; Ahern et al. 2006; Ramin & McMichael 2009; Syaukat 2011). The agricultural sector in Africa is already overwhelmed, and an FAO (2016) report stated that the increase in climate-related events is of significant concern to the particularly vulnerable agricultural sectors. The report concluded that the agricultural sector absorbed 25% of the total impact of climate-related disasters in developing countries between 2003 and 2013. The 2017 report, on the other hand, commented that since agriculture is dependent on weather and climate, water and land to thrive, it is natural for it to be particularly vulnerable to natural disaster (FAO 2017).

However, the impacts of climate change on agriculture and food security have been extensively studied, but only a handful of literature exists on in-depth studies on the impacts of flooding on agriculture and food security in developing countries. This chapter reviews some related literature on this subject within countries in Asia and Africa.

In Asia, Ian (2009) studied climate change, flooding and food security in South Asia and noted that the most vulnerable groups in terms of food security during flooding were the poor, women and children, and the study concluded that flooding disrupted food production more frequently and severely. He also noted that severe cases of flooding affected arable land, causing low nutritional levels in the poor in Bangladesh and other countries in South Asia. In the same vein, Israel and Briones (2012) studied the impacts of natural disasters on agriculture, food security and natural resources and environment in the Philippines, and their results revealed that floods, typhoons and drought have an insignificant negative impact on rice production at the provincial level, while floods have a positive impact on rice production at the local level but not at the national level.

Similarly, Ramakrishna et al. (2014) examined the impact of flood on food security and livelihoods of internally displaced person (IDP) households using a binary logit model. Their findings show that floods negatively impact food security by reducing purchasing power and wage income, causing food shortages among households in the Khammam region of India. The extreme floods of Bangladesh and Pakistan in 2007 and 2010 have been used as case studies by Pacetti et al. (2017) to describe how floods hamper agriculture and food security in developing countries where people are predominantly reliant on agriculture. A combination of remote sensing data, water footprint databases and agricultural statistics were used to examine the impacts of flood events on food supply (food availability). They assessed flood damage on agriculture by estimating crop losses and later converted it into water footprint and lost calories, and their results showed a flood-induced reduction of food supply between 8% and 5% in Pakistan and Bangladesh, respectively. Moreover, the work by Oskorouchi and Sousa-Poza (2021) concentrated on the long-term effect of floods on food security as measured by calorie

and micronutrient consumption in Afghanistan. They found that persons exposed to flooding within a 12-month period had an approximately 60 kcal decreased daily calorie consumption.

In Africa, Zakari et al. (2014) employed a logit model to show the relationship between flooding and food security, and their results show that floods have a negative and significant impact on household food security in Niger Republic, while Oluoko-Odingo (2006) found that a negative correlation existed between flooding and food crop production at the household level in the Nyando district of Kenya.

Nigeria recorded the most devastating flood events between August and October 2012 that submerged hundreds kilometres of rural and urban lands (Ojigi et al. 2013). UN-OCHA (2012) described this 2012 flood as the worst flood recorded in the past 40 years in Nigeria because over 7,705,378 Nigerians were affected, with 2,157,419 IDPs. Farmlands were massively destroyed within the 2012 period, causing food insecurity in parts of Nigeria (FEWS NET 2012, 2013). Similarly, Sidi (2012) found a significant negative impact of the Nigerian 2012 flood on yam, cassava and potato production areas to the tune of 27.9%, 21.6% and 17.2%, respectively. Ajaero (2017) also noted that the 2012 flood affected the food security status of both female- and male-headed households in the country. In addition, Akukwe et al. (2018) studied the effect of flooding on food security in agricultural communities using principal component analysis, and their findings show that floods adversely affect food security in three areas: food supply and distribution, farm labour and facilities and household income and investment.

Furthermore, Jonathan et al. (2020), using a multistage sampling technique, food security index and logistic regression model, did an economic analysis of flood disasters on food security on arable farm households in the southern guinea savanna zone of Nigeria, and their findings revealed that households that experienced flooding were food insecure.

From the foregoing, it is evident that not many substantial studies have been carried out on the impact of floods on agriculture and food security in developing countries, and the few studies done on this concentrated more on the effects on rural dwellers than urban areas. The empirical evidence of this chapter will contribute to the body of literature on impacts of flooding on agriculture and food security because the study area comprised households in both rural and semi-urban areas.

Conceptual framework

The conceptual framework shown in Figure 3.1 illustrates the relationship between flooding and food security formed predominantly from the Food and Agricultural Organization–Food Insecurity and Vulnerability Information Management Systems (FAO-FIVIMS) framework and sustainable livelihood framework as well as reviewed literature. The frameworks aided the understanding of the connections between the four pillars of food security and their influencing factors at different levels (Verduijn 2005; FAO 2008a, 2008b; FAO/NRCB 2008). Figure 3.1 shows that food security has four pillars: food availability, accessibility, utilization and stability. A state of food security implies that an adequate level of food consumption and good nutrition are sustainable at low risk. Stability stresses the importance of minimizing the risk of the negative effects on the other three pillars (FAO 2008a). Conversely, food insecurity exists whenever there is any negative disturbance or shift (e.g. caused by drought or flooding) in any of the pillars of food security, hence the concept of vulnerability.

Vulnerability refers to those factors that induce the risk of households being affected by flooding or becoming food insecure. Vulnerability degrees are determined by the risk factors that households are exposed to and their ability (IPCC 2007, p. 11) to cope with flooding/food insecurity as well as the degree to which they are affected (sensitivity).

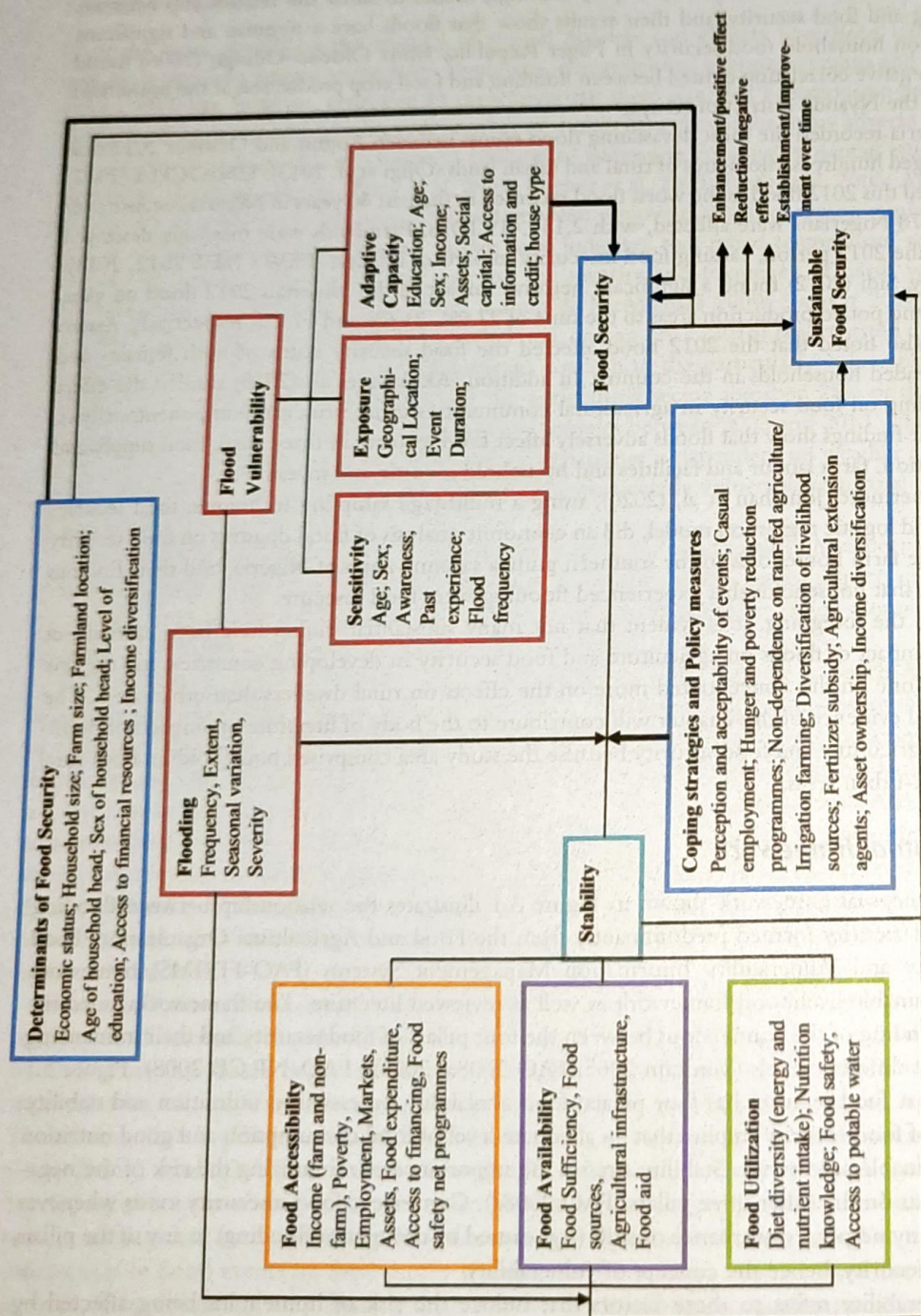


Figure 3.1 Flood, vulnerability and food security framework

Source: Akukwe 2019

On the one hand, flooding is a phenomenon that has a negative impact on food security through reducing labour demand, reducing crop harvest, damaging assets (e.g. houses), reducing farm income derived from crop sales, destroying food/farm storage facilities, destroying road and farmlands and polluting streams that are major water sources. On the other hand, adaptive capacity and coping strategies adopted by households would cushion the effects of flood-induced food insecurity and help improve household food security. Figure 3.1 also shows that food security determinants such as location of farmland and income diversification could either affect or improve food security. In addition, sustainable food security is achieved when the four pillars of food security and coping strategies are maintained over time.

Food security is conceptualized as the dependent variable, and its predictors were drawn from food availability, accessibility, utilization and stability variables measured through the use of indicators (via questionnaire). The categorical variables with an ordinal scale of measurement, such as level of education, classified as no formal education, primary school leaving certificate, secondary school certificate and university degree, were transposed into "no formal education and educated", showing a binary relationship (Akukwe 2019, p. 28).

Research materials and methods

Sample size and sampling method

The study area, southeastern Nigeria, comprises the five states: Anambra, Abia, Enugu, Ebonyi and Imo, located between latitudes 4° 20' to 7° 10' north of the equator and longitudes 6° 35' to 8° 25' east of the Greenwich Meridian, with a land size of about 28,983 km² (Figure 3.2). Southeastern Nigeria experiences the tropical wet-and-dry climate or Aw climate with an average of eight months (March to October) of rainy season and four months of dry season (November to February). Imo and Anambra states were selected for this chapter, as shown in Figure 3.3. Two local government areas (LGAs) that suffer perennial flooding were selected in each state. For equal representation, the Ogbaru and Anambra East LGAs of Anambra state and Ohaji/Egbema and Oguta LGAs of Imo state were purposively selected. According to the National Population Commission (2010), Anambra East has a population of 152,149, Ogbaru has 223,317 persons, Oguta has 142,340 persons and Ohaji/Egbema has 182,891 persons. The sample size was calculated using Yamane's (1967) equation, and according to the equation, any population of 100,000 and more persons would have a sample size of at least 400:

$$n = N + [1 + N(e^2)] \quad (1)$$

where

n – is the sample size

N – is the population of the four LGAs

e – is the level of precision/sampling error, 0.05

n – 400 households

Two communities (including LGA headquarters, which in most cases are semi-urban areas in Nigeria) were selected in each LGA, giving a total of eight communities (four communities for each state) using a multistage purposive sampling technique. Stratified sampling was employed to determine the number of sampled households at the community and LGA levels, while 400 copies of the structured questionnaire were administered through random sampling.

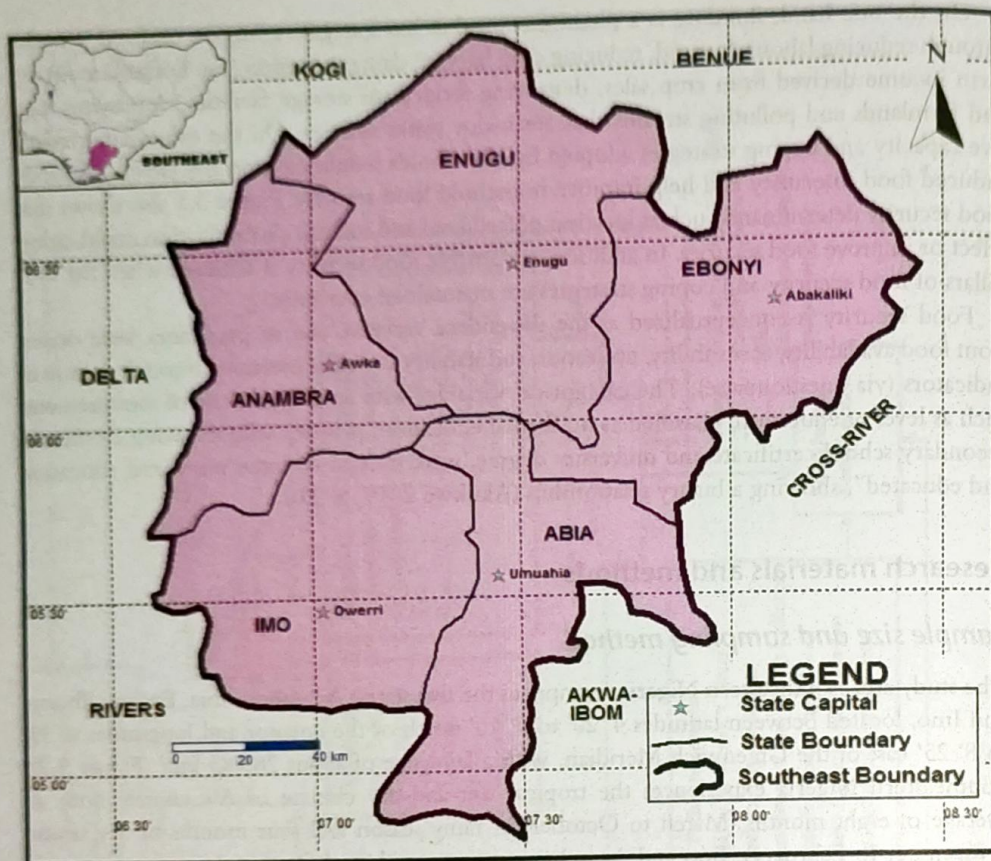


Figure 3.2 Map of the study area

Source: GIS Lab, Department of Geography, University of Nigeria, Nsukka, 2016

Data collection and data analysis

The study was questionnaire based and involved mainly household heads with flood experience between 2012 and 2017. Household food security was calculated using the Household Food Security Survey Module (HFSSM), developed by the USDA. The HFSSM is a set of 18 questions regarding households' food needs (Figure 3.4), and the responses were coded and analysed into food security indices using Rasch analysis. The 18 questions captured the four pillars of food security. The questions had three response categories: often true, sometimes true and never true, in which often true and sometimes true were coded as 1 because they were affirmative responses, and never true was coded as 0 because it depicted a condition of non-occurrence. The scale measures different food security conditions and food insecurity severity ranging from 0 to 10 (Figure 3.5), where 0 represents food secure-households with none or few of the measured conditions, while 10 signifies the most severe condition of food insecurity (Bickel et al. 2000). Household food security status scores were calculated using Rasch analysis, and households were further categorized into four types: food secure, food insecure without hunger, moderately food insecure with hunger and severely food insecure with hunger (Figure 3.5). On the one hand, food-secure households are known to have experienced no or very limited

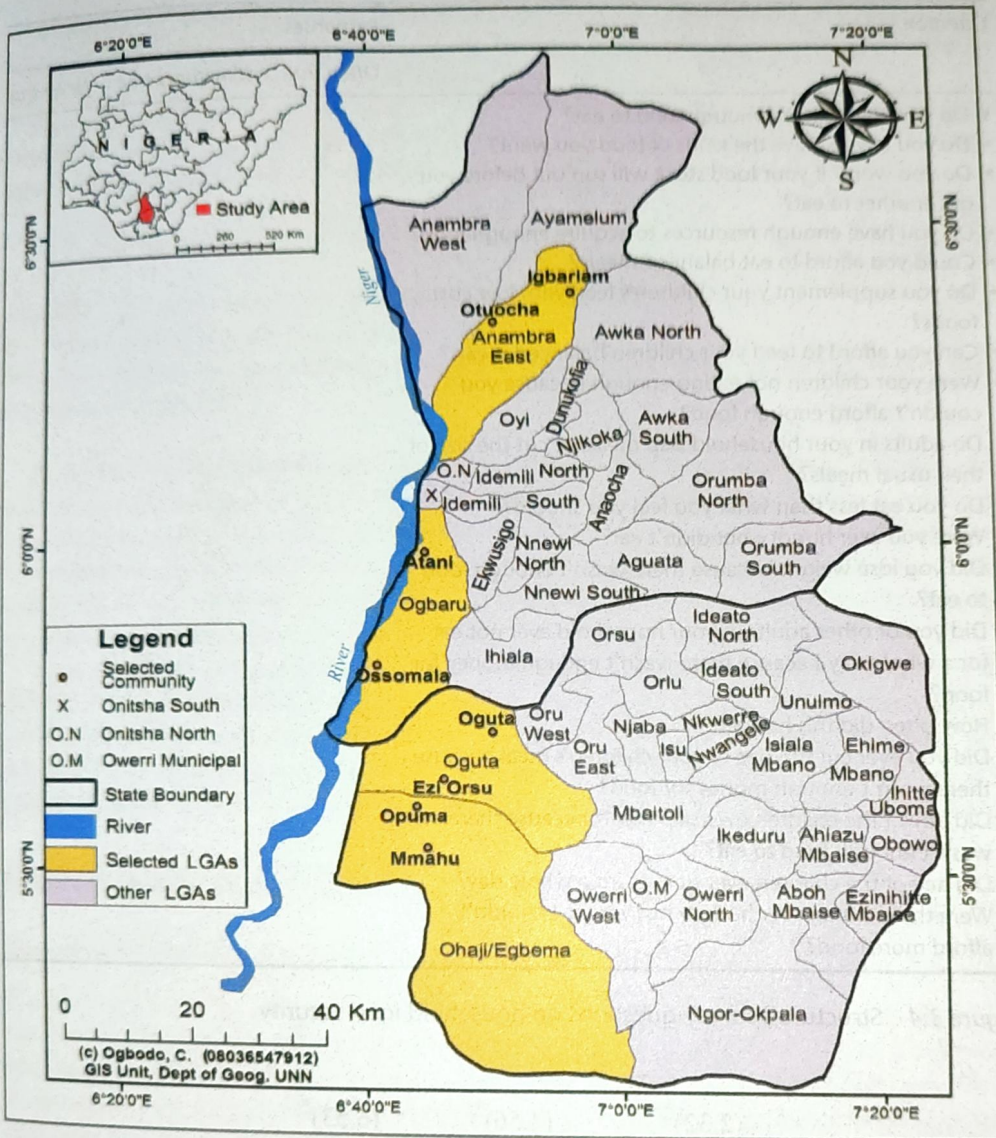


Figure 3.3 Map of the study area showing the sampled LGAs/communities

Source: GIS Lab, Department of Geography, University of Nigeria, Nsukka, 2016

hunger or food insecurity, whereas food-insecure without hunger households either have no or little reduction in food intake without their members experiencing hunger. On the other hand, moderately food-insecure households with hunger had their adults' (with exception of children) food intake reduced, and they experienced hunger repeatedly, whereas severely food-insecure households with hunger repeatedly reduced food intake for both children and adults, and all members experienced hunger (Bickel et al. 2000; Ibok et al. 2014).

Data were collected through a structured questionnaire where the respondents filled out two sets of 18 questions that captured their food security situations before and after flooding, and inferences were made from their responses, majorly extracted from the work of Akukwe (2019).

Question	Responses		
	Often true	Sometimes true	Never true
• Do you always have enough food to eat? • Do you always have the kinds of food you want? • Do you worry if your food stock will run out before you get another to eat? • Do you have enough resources to acquire enough food? • Could you afford to eat balanced meals? • Do you supplement your children's feed with low cost foods? • Can you afford to feed your children balanced meals? • Were your children not eating enough because you couldn't afford enough food? • Do adults in your household skip meals or cut the size of their usual meals? • Do you eat less than what you feel you should? • Were you ever hungry but didn't eat? • Did you lose weight because there wasn't enough food to eat? • Did you or other adults in your household ever not eat for a whole day because there wasn't enough money for food? • How often did this happen? • Did you ever cut the size of your children's meal because there wasn't enough money for food? • Did any of the children ever skip meals because there wasn't enough food to eat? • Did any of the children ever not eat for a whole day? • Were the children ever hungry but you just couldn't afford more food?			

Figure 3.4 Structured survey questions on household food security

	(2.32) ^{*/}	(4.56) ^{*/}	(6.53) ^{*/}
Food Secure	Food Insecure:		
	Food Insecure Without Hunger	Food Insecure With Hunger:	
		(less severe) "Moderate"	(more severe) "Severe"

Figure 3.5 Household food security status – categorical measure

Source: Bickel et al. 2000, p. 31

*/ located at midpoint between the two adjacent household scale values.

The extent to which flood negatively affected agriculture and food security was measured on a 5-point Likert scale, with 5 – to a very great extent, 4 – to a great extent, 3 – to a moderate extent, 2 – to little extent and 1 – to no extent.

In order to establish the nature of the relationship that exists between floods and food security status, the determinants of household food security were assessed using multiple binary logistic regression, as seen in the equation

$$[Y / (1 - Y)] = b_0 + b_1x_1 + b_2x_2 + \dots b_{25}x_{25} \quad (2)$$

where $[Y / (1 - Y)]$ – likelihood that household is food secure/insecure

b_0 – constant that scales the equation

b_1 to b_{25} – coefficients that determine the direction and extent to which the (x) factors affect food security status (Y)

x_1 to x_{25} – factors that influence food security: age, sex of household head, marital status, level of education, literacy rate, diversified income, flood experience, monthly income, off-farm income, dependency ratio, private land ownership, group membership, sufficient own food production, poultry farm ownership, livestock ownership, distance to farm, storage facility availability, distance to market, fertilizer use, food/aid receipt, irrigation, nutrition knowledge, financial support, farm size and access to credit.

Multiple binary logistic regression was selected because of the dichotomous nature of the dependent variable (food secure/food insecure). To show the relationship between food security and its determinants using a binary model, the classified HFSSM food security levels of four were transposed into two: food secure and food insecure (consisting of severely food insecure with hunger, moderately food insecure with hunger and food insecure without hunger) households.

Research results and discussion

Household food security status before and after flooding

The food security status of households before and after flood events is shown in Table 3.1. It was observed that preceding flood disasters, 33.3% of households were food secure, 40.2% were food insecure without hunger, 13% were moderately food insecure with hunger and 13.5% were severely food insecure with hunger in the study area. This indicates that a third of the total population had no food insecurity problem; 40.2% had no or little reduction in food intake; 13% had their adult members' food intake reduced, and they repeatedly experienced hunger; and 13.5% had both their children and adults' food intake reduced, and they repeatedly experienced hunger.

Table 3.1 also reveals that after flooding, there was a negative shift in household food security levels, as the number of food-secure households decreased from 33.3% to 7.2%, causing a 26.1% reduction in food-secure households. Reduced food intake for both adults and children resulting from inadequate resources to acquire food and reduced household food supply caused the extremely higher number of food-insecure households recorded after flooding.

However after flooding, food-insecure without hunger households reportedly reduced to 39.3% from the 40.2% that was recorded before flooding. The table also shows that flooding induced a greater percentage of households into being severely food insecure, with a 2.7% increase in the number of moderately food-insecure with hunger households and a 24.3% increase in severely food-insecure with hunger households. Our findings reveal that flooding affected food

Table 3.1 Before- and after-flood differential household food security status/levels in southeastern Nigeria

Community	Before-flood category of food security status/level (%)				Total
	Food secure	Food insecure without hunger	Moderately food insecure with hunger	Severely food insecure with hunger	
Atani	37.5	37.5	9.4	15.6	64
Ossomala	25.0	45.3	20.3	9.4	64
Otuocha	43.2	40.9	13.6	2.3	44
Igbariam	25.6	34.9	20.9	18.6	43
Oguta	26.8	43.9	9.8	19.5	41
Ezi-Orsu	30.0	35.0	20.0	10.0	40
Mmahu	44.2	34.6	3.9	17.3	52
Opuoma	32.8	48.1	3.9	15.4	52
Total (mean)	33.3	40.2	13.0	13.5	400

Community	After flood category of food security status/level (in percent %)				Total
	Food secure	Food insecure without hunger	Moderately food insecure with hunger	Severely food insecure with hunger	
Atani	6.2	39.1	14.1	40.6	64
Ossomala	1.6	45.3	12.5	40.6	64
Otuocha	9.1	59.1	20.4	11.4	44
Igbariam	0.0	20.9	7.0	72.1	43
Oguta	22.0	19.5	43.9	14.6	41
Ezi-Orsu	7.5	40.0	12.5	40.0	40
Mmahu	7.7	40.4	13.4	38.5	52
Opuoma	7.7	44.2	7.7	40.4	52
Total (mean)	7.2	39.3	15.7	37.8	400

Source: Akukwe (2019)

security by generally causing an increase in the number of food-insecure households to 92.8%, a 26.5% increase over households that were food insecure before flooding in southeastern Nigeria.

The Igbariam community was the most affected after flooding, as 100% of the households became food insecure, where 72.1% became severely food insecure with hunger. The Ossomala community was the second most affected, as after the flooding events, only 1.6% of households remained food secure. After flooding, the Atani, Mmahu, Ezi-Orsu and Opuoma communities also were affected, as fewer than 8% of their households were food secure, revealing that more than 90% of their households experienced food insecurity after flooding. Flooding affected the Oguta community least, because 22% of its households remained food secure after flooding, showing a 4.8% reduction in the number recorded before flooding. Thus, flooding caused a shift that increased the number of *food insecurity hotspots* because most households (except in the Oguta community) moved from food insecure without hunger to either moderately or severely food insecure with hunger after flooding. This implies that a greater number of households will need food assistance to improve their coping ability with food insecurity induced by flooding since a majority of the households repeatedly reduced children's and/or adults' food intake after a flood experience. Atani, Otuocha, Mmahu and Oguta are LGA headquarters and are considered semi-urban areas. Our findings show that floods affect household food security of agrarian communities in both rural and urban areas equally (Akukwe 2019, pp. 155–156).

The negative impacts of flooding on agriculture and food security

Table 3.2 shows the areas and extent to which flooding negatively affected food security and agriculture. Our findings reveal that the impacts of flooding on food security and agriculture were to a great extent as measured on the Likert scale in Table 3.2. The largest percentage of households reported that flooding caused a reduction in farm income derived from crop sales (55.5%), reduction in crop harvest (55.7%), destruction of roads (56%) and increase in food prices (63.5%). Floods were reportedly found to reduce labour demand, affect the quality of food eaten, pollute streams, destroy food/farm storage facilities and reduce the number of times food was consumed and the quantity of food eaten to a great extent, as indicated by 25.3%, 37%, 40.3%, 43.5%, 46.8%, and 47.5% of households respectively. However, 52.5% reported flooding to have caused stream pollution to a very great extent, which affected communities greatly since streams are their major source of water. Our findings are in agreement with those of Devereux (2007), Odufuwa et al. (2012), Ajaero and Mozie (2014), Otomofa et al. (2015), Ikani (2016) and Oskorouchi and Sousa-Poza (2021). Ajaero and Mozie (2014) revealed that flooding caused stream pollution in Ogbaru and Anambra East LGAs, and Ikani (2016) noted loss of farm crops due to flooding in the Gwagwalada area in Nigeria, while Oskorouchi and Sousa-Poza (2021) found flood experiences to be associated with lower diet quantity and quality in Afghanistan.

Flooding was revealed to cause changes in food consumption patterns arising from destruction of farmlands, rise in food prices, disruption of sources of livelihoods, pollution of streams, reduction in crop harvest, decrease in quality and quantity of food eaten, destruction of roads and reduction in meal frequency, which adversely affected food accessibility, availability and utilization over time.

This chapter has revealed that flooding negatively affects agriculture and food security, and this could hamper the achievement of Sustainable Development Goal 2, with an emphasis on achieving food security and promoting sustainable agriculture; thus adequate flood management is necessary in southeastern Nigeria.

Table 3.2 Negative effects of flooding on agriculture and food security

Negative effect of flooding on food security	Extent of effect				
	1	2	3	4	5
Reduces crop harvest	0.0	1.3	2.3	55.7	40.7
Decreases farm income derived from crop sales	0.0	2.5	4.0	55.5	38.0
Destroys road	0.0	10.3	18.0	56.0	15.7
Destroy food/farm storage facilities	0.0	5.0	29.3	43.5	22.2
Reduces labour demand	3.3	24.0	25.7	25.3	21.7
Pollutes streams	0.0	2.5	4.7	40.3	52.5
Reduces the number of times food is consumed	0.0	16.7	27.5	46.8	9.0
Affects the quality of food eaten	6.0	18.5	26.5	37.0	12.0
Affects the quantity of food eaten	0.0	16.0	31.0	47.5	5.5
Increases food item prices	0.0	0.5	19.7	63.5	16.3

Source: Field work, 2017

A 5-point scale with 5 = to a very great extent, 4 = to a great extent, 3 = to a moderate extent, 2 = to little extent, 1 = to no extent.

Relationship between flooding and food security

Table 3.3 shows the 25 variables that predicted food security status in southeastern Nigeria. These variables range from demographic and socio-economic factors to environmental factors, but because this chapter is centred on impacts of flooding on agriculture and food security, only flood experience as a predictor is discussed.

The coefficient of the relationship between flood experience and food security was negative (-1.110) and statistically significant at 5%. This implies that the more a household experiences flooding, the more food insecure that household becomes. This also means that for every unit increase in flood experience, the odds ratio of the household being food secure reduces by a factor of 0.33 (Table 3.3). Our findings agree with those of Ramakrishna et al. (2014) and Zakari et al. (2014), who found floods to have a negative impact on household food security in Khammam (India) and Niger, respectively. The negative effects of floods have been discussed in the preceding pages, and they include destruction of roads leading to farmlands; destruction of farmland and storage facilities, which leads to food shortage and crop failure; reduction in crop

Table 3.3 Logit model output of determinants of food security status

Factors influencing food security	B	S.E.	Wald	df	Sig.	Exp(B)
Sex of household head	.945	.433	4.763	1	.029*	2.572
Age of household head	-.128	.163	.615	1	.433	.880
Marital status	.381	.142	7.239	1	.007*	1.464
Literacy rate	-.616	.581	1.124	1	.289	.540
Level of education	.502	.260	3.715	1	.054*	1.652
Diversified income	.206	.854	.058	1	.810	1.228
Off-farm income	1.432	.679	4.454	1	.035*	4.189
Monthly income	.680	.192	12.533	1	<.0001*	1.973
Dependency ratio	.879	.389	5.115	1	.024*	2.410
Group membership	-.197	.428	.211	1	.646	.822
Private land ownership	-.455	.454	1.004	1	.316	.635
Sufficiency in own food production	1.068	.430	6.175	1	.013*	2.908
Livestock ownership	.893	.410	4.743	1	.029*	2.443
Village/agric. poultry ownership	.839	.386	4.711	1	.030*	2.314
Distance to farm	-.530	.434	1.489	1	.222	.589
Distance to market	.373	.382	.953	1	.329	1.451
Storage facility availability	-.019	.383	.003	1	.959	.981
Fertilizer use	.564	.373	2.293	1	.130	1.758
Irrigation practice	1.997	.659	9.193	1	.002*	7.367
Food/aid receipt	.333	.397	.705	1	.401	1.395
Nutritional knowledge	.807	.536	2.265	1	.132	2.241
Farm size	.036	.300	.014	1	.906	1.036
Financial support	.328	.391	.704	1	.401	1.388
Credit access	-.540	.585	.850	1	.356	.583
Flood experience	-1.110	.337	10.868	1	.001*	.330
Constant	-7.305	1.555	22.084	1	.000	.001
-2 Log likelihood	286.408					
Cox and Snell R ²	.426					
Nagelkerke R ²	.592					

* Significant at 0.05 level of significance ($P \leq 0.05$).

harvest; destruction of roads; and reduction in meal frequency, food quality and food quantity, thereby increasing food insecurity.

As shown in the conceptual framework, the negative impacts of flooding on agriculture and food security could be minimized by putting in place some coping strategies and policy measures such as mechanized and irrigation farming to encourage all-year-round planting and harvest and food safety net and poverty alleviation programmes to help the poor, who are the most vulnerable to the impacts of flooding, as supported by Pacetti et al. (2017).

Conclusion

This chapter analysed the impacts of flooding on agriculture and food security in developing countries. Flooding is an intractable environmental problem that has negative impacts on agriculture and food security in developing countries because their major sources of livelihood are affected. Flooding caused a negative shift in household food security levels by causing an increase in the number of food-insecure households to 92.8%, showing a 26.5% increase over households that were food insecure before flooding in southeastern Nigeria. Reduced food intake for both adults and children resulted from inadequate resources to acquire food and reduced household food supply, causing the high percentage of severely food-insecure households recorded after flooding. Our findings also revealed that floods affect household food security of agrarian communities in both rural and urban areas equally.

Flooding was revealed to cause changes in food consumption patterns arising from destruction of farmlands, rise in food prices, disruption of sources of livelihoods, pollution of streams, reduction in crop harvest, decrease in quality and quantity of food eaten, destruction of roads and reduction in meal frequency, which significantly affected food accessibility, availability and utilization over time. The coefficient of the relationship between flood experience and food security was negative (-1.110) and statistically significant at 5%, with an odds ratio of 0.33. This implies that the more a household experiences flooding, the more food insecure that household becomes. This also means that for every unit increase in flood experience, the odds ratio of the household being food secure reduces by a factor of 0.33.

Consequently, since our findings show that flooding has negative impacts on agriculture and food security, this would hamper the attainment of Sustainable Development Goal 2, which emphasizes achieving food security and promoting sustainable agriculture; thus adequate flood management and policy options geared towards enhancing food security are pertinent in southeastern Nigeria.

Recommendations

1. Irrigation farming to encourage all-year-round and off-season planting, which will boost crop productivity and alter the harvest season, especially for crops like yams that are usually harvested during flooding periods (July–October), thereby reducing food wastage.
2. Food safety net and poverty alleviation programmes to help the poor who are the most vulnerable to the impacts of flooding through making food available and accessible.

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